Conservation Laws:

Material Derivative:

Velocity Field:

PAC-NeRF: Physical Constraints

Conservation Laws:

Density Field

Color Field

$$\frac{D\sigma}{Dt} = 0 \qquad \frac{D\mathbf{c}}{Dt} = 0$$

Material Derivative:

$$\frac{D\varphi}{Dt} = \frac{\partial\varphi}{\partial t} + \mathbf{v} \cdot \nabla\varphi$$

Velocity Field:

Cauchy Stress Tensor

$$\rho\frac{D\mathbf{v}}{Dt} = \nabla\mathbf{T} + \rho\mathbf{g}$$

PAC-NeRF: Governing Equations

$\mathbf{F}$ = Deformation gradient

$\mathbf{T}$ = Stress Tensor

$$J\mathbf{F} = \det(F)$$

Material	Cauchy Stress Tensor
Elasticity	$J\mathbf{T}(\mathbf{F}) = \mu(\mathbf{F}\mathbf{F}^T) + (\lambda \log(J) - \mu)\mathbf{I}$
Newtonian Fluid	$J\mathbf{T}(\mathbf{F}) = \frac{1}{2}\mu\left(\nabla\mathbf{v} + \mathbf{v}^T\right) + \kappa\left(J - \frac{1}{J^6}\right)$
Plasticine	$J\mathbf{T}(\mathbf{F}) = \mathbf{U}\left(2\mu\epsilon + \lambda Tr(\epsilon)\right)\mathbf{U}^T$